ApproxDA-TransUNet: A Lightweight Dual-Attention Architecture for Medical Image Segmentation

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Abstract

Dual-attention networks such as DA-TransUNet model spatial (position) and channel dependencies on top of a TransUNet backbone, at a cost of ≈108M parameters and ≈30 GFLOPs per 224×224 slice. We present **ApproxDA-TransUNet**, a lightweight dual-attention segmentation network that combines a ResNet-50 stem truncated at stride 8, a LightViT-T bottleneck transformer, approximated dual-attention blocks on the bottleneck and all three skip connections, and a slim decoder. In each block, position attention is restricted to local windows ($M = 28$) and projected to a low rank ($r = 32$), and channel attention is computed in groups ($G = 8$). The model uses 90.0% fewer parameters (10.75M vs. 107.95M) and 77.1% fewer FLOPs (6.92 vs. 30.21 GFLOPs) than DA-TransUNet. In single-axis ablations on Synapse, removing the ApproxDA blocks reduces Dice by 2.76 points ($p = 0.008$), and the approximation costs no accuracy: exact global dual attention, as in DA-TransUNet, gives the same Dice (-0.01 points) at 14% more FLOPs and twice the training time. Across five benchmarks (Synapse CT, ACDC MRI, CVC-ClinicDB colonoscopy, ISIC 2018 dermoscopy, and FeTA 2.4 fetal MRI) with three seeds each, ApproxDA-TransUNet retains the Dice of DA-TransUNet: its three-seed mean is higher on all five datasets (average gain +0.37 points, 10 of 15 seed pairs positive), although no single dataset shows a significant difference and HD95 is higher on four of the five datasets.

Keywords: Biomedical image segmentation, Lightweight architecture, Dual-attention approximation, Vision Transformer, Multi-organ

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1. Introduction

Automated biomedical image segmentation is fundamental to computer-assisted diagnosis, radiotherapy planning, and surgical navigation [1, 2, 3]. Clinical modalities span a broad spectrum—from 3D volumetric CT and MRI to 2D endoscopic video and high-resolution dermoscopy. Accurate boundary delineation across these domains requires architectures capable of both fine-grained edge localization and expansive global context modeling.

While convolutional neural networks (CNNs) [1, 2, 4] form the conventional baseline, their localized receptive fields limit long-range context aggregation. Hybrid CNN-Transformer frameworks like TransUNet [3] address this via self-attention, and dual-attention networks [5] concurrently capture spatial affinities via Position Attention Modules (PAM) and cross-channel correlations via Channel Attention Modules (CAM). Recently, DA-TransUNet [6] integrated dual-attention blocks across multi-scale skip connections and the bottleneck, demonstrating state-of-the-art accuracy on multi-organ CT and colon polyp benchmarks.

Despite their representational strength, standard dual-attention architectures suffer from severe computational overhead. Full-map PAM scales quadratically with spatial tokens ($\mathcal{O}(N^2C)$), while CAM scales quadratically with channel depth ($\mathcal{O}(C^2N)$). Coupled with a ViT-B/16 encoder and a wide decoder, DA-TransUNet requires **107.95M parameters** and **30.2 GFLOPs** per 224×224 slice (39.8 GFLOPs at 256×256), which limits its use on resource-constrained hardware.

In this work, expanding upon our preliminary workshop study [7], we present **ApproxDA-TransUNet**, a system-level redesign of DA-TransUNet built from four components: (i) a *ResNet-50 stem truncated after its second stage* (stride 8, 512 channels) followed by a compact Vision Transformer (*LightViT-T* [8]), which reduces the encoder from 97.9M to 9.3M parameters; (ii) *approximated dual-attention blocks*, in which position attention is restricted to local windows ($M = 28$) and projected to a low rank ($r = 32$), channel attention is computed in groups ($G = 8$), and the two branches are blended with a fixed gate ($g = \frac{1}{2}$); (iii) these blocks on all three skip connections (1/8, 1/4, 1/2) and on the bottleneck, as in DA-TransUNet; and (iv) a *slim decoder* with additive skip fusion and 64 channels at every skip stage.

Together, these changes reduce the parameters from **107.95M to 10.75M** (-90.0%) and the compute from **30.21 to 6.92 GFLOPs** (-77.1%).

In single-axis ablations on Synapse, removing the dual-attention blocks reduces Dice by 2.76 points, and their approximation costs no accuracy: replacing the approximated attention by the exact global dual attention of DA-TransUNet leaves Dice unchanged while increasing FLOPs by 14% and doubling the training time.

We evaluate ApproxDA-TransUNet across five multi-modal benchmarks under a three-seed protocol (1234, 0, 1): Synapse CT, ACDC cardiac MRI, CVC-ClinicDB colonoscopy, ISIC 2018 dermoscopy, and FeTA 2.4 fetal brain MRI. ApproxDA-TransUNet retains the Dice of DA-TransUNet: its three-seed mean Dice is higher on all five datasets (average gain +0.37 percentage points, 10 of 15 seed pairs positive), although no single dataset shows a significant difference, and HD95 is higher than DA-TransUNet's on four of the five datasets. On FeTA 2.4, the 10.75M model is 1.8–2.5 points above the reproduced U-Net family and Swin-UNet and within 0.1 points of TransUNet.

The primary contributions of this paper are:

- **Lightweight dual-attention architecture**: ApproxDA-TransUNet uses 90% fewer parameters and 77% fewer FLOPs than DA-TransUNet at comparable segmentation accuracy.
- **Dual-attention approximation**: windowed low-rank position attention (M, r) and grouped channel attention (G), with single-axis ablations on Synapse showing that removing the blocks reduces Dice by 2.76 points and that the approximation matches exact dual attention at lower cost.
- **Five-dataset, three-seed validation**: paired comparisons with DA-TransUNet on benchmarks spanning CT, MRI, colonoscopy, dermoscopy, and fetal MRI, with literature and reproduced baselines and a deployment profile.

The rest of this paper is organized as follows: Section 2 reviews related work; Section 3 presents the methodology; Section 4 reports experimental results; Section 5 provides ablation studies; Section 6 discusses augmentation, deployment cost, and limitations; Section 7 concludes the paper.

2. Related Work

2.1. Architectures for Biomedical Image Segmentation

U-Net [1] established the dominant encoder-decoder topology for medical image segmentation, using multi-scale skip connections to preserve high-resolution spatial details. Subsequent variants enhanced representation capacity through nested dense skips (UNet++ [2]), additive attention gates (Attention U-Net [4]), and multi-resolution residual blocks (MultiResUNet [9]). However, purely convolutional models remain constrained by localized receptive fields, hindering the disambiguation of contiguous anatomical structures.

To overcome this, Vision Transformers (ViTs) [10] were adopted for global context aggregation. TransUNet [3] pioneered hybrid architectures by coupling a ResNet-50 CNN with a ViT-B/16 bottleneck transformer. Swin-UNet [11] explored a purely transformer-based U-shaped structure with shifted local windows, while UNETR [12] and SwinUNETR [13] extended transformers to volumetric 3D segmentation. While effective, these models often exceed 100M parameters, imposing prohibitive memory and computational overhead.

2.2. Dual-Attention Mechanisms and Computational Bottlenecks

Dual attention was formalized by Fu et al. [5] (DANet) to simultaneously model spatial dependencies via Position Attention Modules (PAM) and cross-channel interdependencies via Channel Attention Modules (CAM). In biomedical imaging, DA-TransUNet [6] embedded dual-attention blocks across the bottleneck and skip connections of TransUNet, establishing state-of-the-art results on multi-organ CT and polyp segmentation. Related efforts explored dual attention in hierarchical transformers (DAE-Former [14], TransDAE [15]) and state-space architectures (DAMamba [16]). Contextual attention networks such as SECA-Net [17] add squeezed-and-excitated multi-scale atrous convolution and spatial-channel attention modules to the bottleneck of a ResNet-based U-Net to strengthen context modeling. Our work takes a complementary, efficiency-oriented view: we keep the dual-attention formulation and show that its spatial and channel attention can be approximated aggressively without a measurable loss in accuracy (Section 5).

However, standard dual attention exhibits quadratic complexity: spatial self-attention scales as $\mathcal{O}(N^2C)$ for $N = H \times W$ tokens, while channel attention scales as $\mathcal{O}(C^2N)$. In multi-scale architectures, generating full affinity

matrices across multiple high-resolution skip stages (112×112 and 56×56) makes memory grow quadratically with the number of tokens.

2.3. Efficient Attention Approximation and Positioning

To mitigate quadratic complexity, diverse approximation paradigms have emerged in general vision: local window partitioning (Swin Transformer [18]), low-rank key-value projection (Linformer [19], Performer [20]), channel group factorizations (EMCAD [21]), and efficient transformer backbones (LightViT [8]).

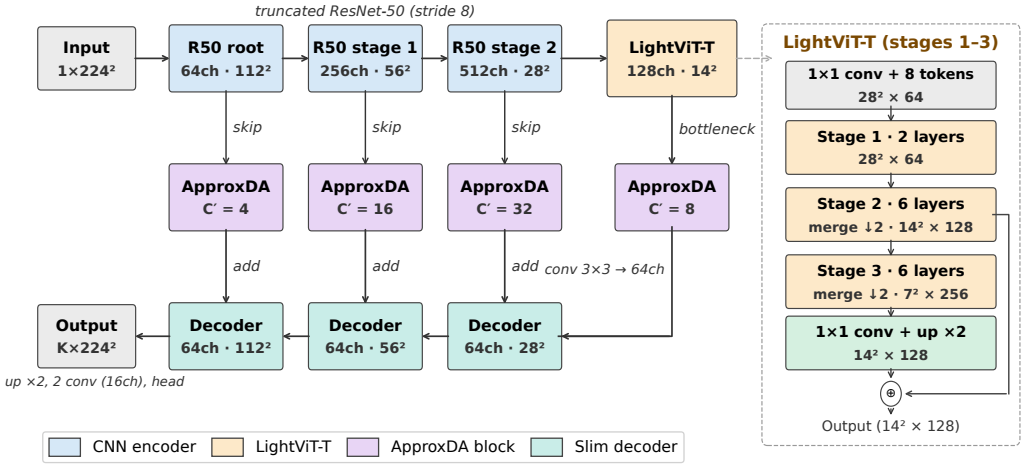
Efficient medical image segmentation has also been approached from other, complementary directions. On the decoder side, Chen et al. [22] select decoder blocks adaptively and reweight filters to bypass redundant blocks and obtain a lightweight model for each dataset. Other recent architectures capture multi-scale or global context at lower cost without full transformer attention; FE-MAANet [23], for example, combines adaptive large-kernel modules built from cascaded depthwise-separable convolutions with a frequency-spatial parallel attention module in the skip connections, and EMCAD [21] designs an efficient multi-scale attention decoder. In contrast, our work starts from a dual-attention Transformer U-Net (DA-TransUNet) and jointly simplifies the backbone, the spatial and channel attention operators, the skip processing, and the decoder within one fixed architecture, without architecture search or auxiliary routing. Our earlier workshop study [7] analyzed window sensitivity within the heavy ViT-B model; this work turns that analysis into a lightweight architecture and shows that a 90% parameter reduction is possible at comparable accuracy.

3. Methodology

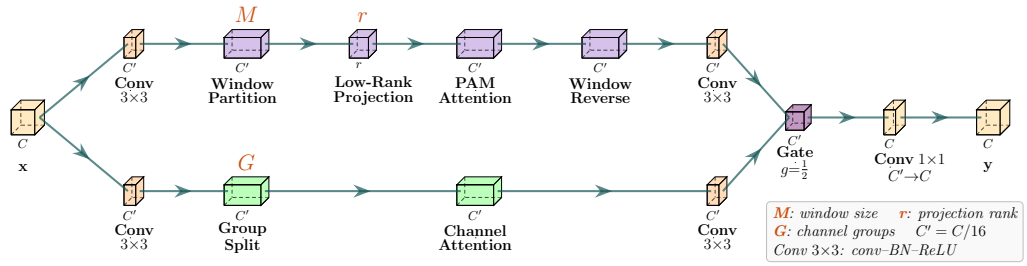
This section describes ApproxDA-TransUNet: the overall topology, the attention approximation in the ApproxDA block, the slim decoder, and the computational cost.

3.1. Overall Lightweight Architecture

Figure 1a shows the complete architecture. Compared with the 108M-parameter DA-TransUNet [6], every stage is redesigned for efficiency while keeping the multi-scale skip structure.



(a) Overall architecture. The encoder consists of a ResNet-50 truncated after its second stage (stride 8) followed by LightViT-T, which produces the $14 \times 14 \times 128$ bottleneck. ApproxDA blocks are placed on all three skip connections (1/8, 1/4, 1/2) and on the bottleneck; C' is the compressed width inside each block. The decoder uses 64 channels at every skip stage and 16 channels at full resolution.



(b) ApproxDA block: a windowed low-rank PAM branch (M, r) and a grouped CAM branch (G), each between two 3×3 convolutions, fused with a fixed gate $g = \frac{1}{2}$.

Figure 1: ApproxDA-TransUNet: (a) the overall architecture (10.75M parameters) and (b) the ApproxDA block.

3.1.1. Early-Truncated CNN Stem

In TransUNet and DA-TransUNet, the hybrid encoder runs the first three stages of ResNet-50 (down to stride 16 and 1024 channels; 11.9M parameters, 4.0 GFLOPs). The resulting 14×14 feature map is projected to 768-dimensional tokens and processed by a 12-layer ViT-B/16 (85.1M parameters, 17.4 GFLOPs).

ApproxDA-TransUNet truncates ResNet-50 after its second stage. For an input image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$ (typically 224×224 or 256×256):

- **Stem & Stage 1**: Generates feature maps $\mathbf{F}_{1/2} \in \mathbb{R}^{64\times \frac{H}{2}\times \frac{W}{2}}$ (stride 2) and $\mathbf{F}_{1/4} \in \mathbb{R}^{256\times \frac{H}{4}\times \frac{W}{4}}$ (stride 4).
- **Stage 2**: Produces feature map $\mathbf{F}_{1/8} \in \mathbb{R}^{512\times \frac{H}{8}\times \frac{W}{8}}$ (stride 8).

The third stage is removed, so the CNN stem has 1.45M parameters (1.83 GFLOPs).

3.1.2. LightViT-T Bottleneck and Attention Placement

Instead of ViT-B, $\mathbf{F}_{1/8}$ is projected by a 1×1 convolution and processed by the three stages of LightViT-T [8] (ImageNet-1k weights), a hierarchical lightweight vision transformer. The second-stage output and the upsampled third-stage output are fused into the bottleneck representation $\mathbf{Z} \in \mathbb{R}^{128\times \frac{H}{16}\times \frac{W}{16}}$ (14×14 for 224×224 input). LightViT-T has 7.81M parameters but only 0.75 GFLOPs, because it operates on 28×28 and smaller token grids. The bottleneck ApproxDA block is positioned post-transformer, directly operating on $\mathbf{Z}$ to refine global representations before decoding.

3.2. The ApproxDA Block Formulation

Figure 1b illustrates the internal topology of the ApproxDA block. The module receives input tensor $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ and splits computation into two parallel streams: Low-Rank Windowed Position Attention (PAM) and Grouped Channel Attention (CAM).

3.2.1. Bottleneck Convolutional Compression

As in the DANet head used by DA-TransUNet, each branch first compresses $\mathbf{X}$ with its own convolution:

$$\mathbf{X}' = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times 3}(\mathbf{X}))) \in \mathbb{R}^{C'\times H\times W}, \tag{1}$$

where $C' = C/16$. For the bottleneck and the 1/8, 1/4, and 1/2 skips, C' is 8, 32, 16, and 4.

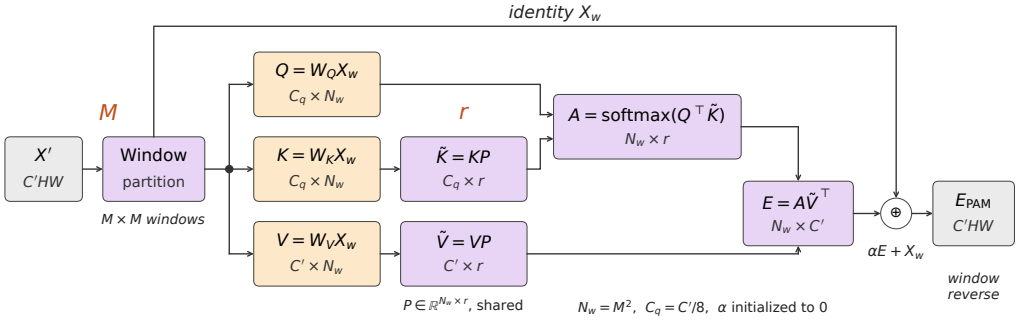


Figure 2: Low-rank windowed position attention, shown for one $M \times M$ window. Keys and values are projected from $N_w = M^2$ tokens to rank r by a projection P shared between them, so the attention matrix A is $N_w \times r$ instead of $N_w \times N_w$; the output is added to the input with the scale α.

3.2.2. Low-Rank Windowed Position Attention (PAM)

Figure 2 shows the computation inside one window. Full spatial self-attention requires pairwise affinities between all $N = HW$ tokens, yielding $\mathcal{O}(N^2 C')$ complexity. We approximate this along two orthogonal dimensions: **spatial windowing** (M) and **low-rank projection** (r).

Window Partitioning. The feature map $\mathbf{X}'$ is partitioned into non-overlapping spatial windows of size $M \times M$, where $M = \min(M, H, W)$ ($M = 28$ for 224×224 inputs). Consequently, the $1/2$ and $1/4$ skip stages are partitioned into 4×4 and 2×2 window grids, while the $1/8$ skip and bottleneck operate within a single global window. Within each window of $N_w = M^2$ tokens, queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ are generated via 1×1 convolutions:

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{X}_w, \quad \mathbf{K} = \mathbf{W}_K \mathbf{X}_w \in \mathbb{R}^{C_q \times N_w}, \quad \mathbf{V} = \mathbf{W}_V \mathbf{X}_w \in \mathbb{R}^{C' \times N_w}, \tag{2}$$

where $C_q = \max(1, \lfloor C'/8 \rfloor)$.

Low-Rank Subspace Projection. Rather than computing an $N_w \times N_w$ attention matrix, keys and values are projected into a subspace of rank r ($r = 32$) via shared projection matrix $\mathbf{P} \in \mathbb{R}^{N_w \times r}$:

$$\tilde{\mathbf{K}} = \mathbf{K}\mathbf{P} \in \mathbb{R}^{C_q \times r}, \quad \tilde{\mathbf{V}} = \mathbf{V}\mathbf{P} \in \mathbb{R}^{C' \times r}. \tag{3}$$

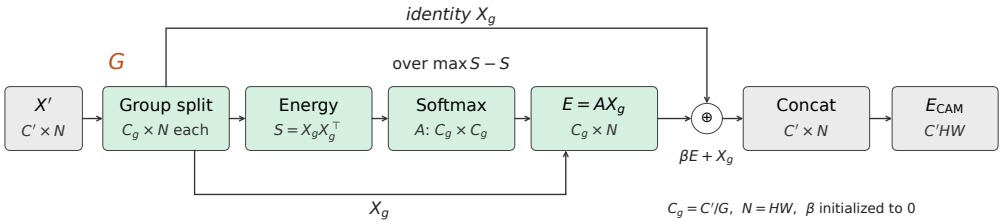


Figure 3: Grouped channel attention, shown for one of the G groups. The channel affinity A is computed within each group of $C_g = C'/G$ channels, the output is added to the input with the scale β, and the groups are concatenated.

The windowed position attention matrix $\mathbf{A} \in \mathbb{R}^{N_w \times r}$ and refined features $\mathbf{E}_{\text{PAM}}$ are then computed as:

$$\mathbf{A} = \text{softmax}\left(\mathbf{Q}^\top \tilde{\mathbf{K}}\right), \tag{4}$$

$$\mathbf{E}_{\text{PAM}} = \alpha \left(\mathbf{A}\tilde{\mathbf{V}}^\top\right)^\top + \mathbf{X}_w \in \mathbb{R}^{C' \times N_w}, \tag{5}$$

where α is a learnable scalar initialized to 0. All window features are reassembled and passed through a 3×3 refinement convolution. Setting $r = 0$ bypasses projection, recovering exact attention within each window.

3.2.3. Grouped Channel Attention (CAM)

Figure 3 shows the computation for one group. Standard CAM computes channel covariance across all C' channels with $\mathcal{O}(C'^2 N)$ complexity. ApproxDA blocks parameterize channel interactions via grouping parameter G: the C' channels are partitioned into G disjoint groups of width $C_g = C'/G$:

$$\mathbf{X}' = \left[\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \ldots, \mathbf{X}^{(G)}\right], \quad \mathbf{X}^{(g)} \in \mathbb{R}^{C_g \times N}. \tag{6}$$

Within each group g, the channel correlation energy between channel slice i and j is $s_{ij} = \mathbf{X}_i^{(g)} \cdot \mathbf{X}_j^{(g)}$. The channel attention map and modulated features are given by:

$$x_{ij}^{(g)} = \text{softmax}_j\left(\max_k s_{ik} - s_{ij}\right), \tag{7}$$

$$\mathbf{E}_{\text{CAM},i}^{(g)} = \beta \sum_{j=1}^{C_g} x_{ij}^{(g)} \mathbf{X}_j^{(g)} + \mathbf{X}_i^{(g)}, \tag{8}$$

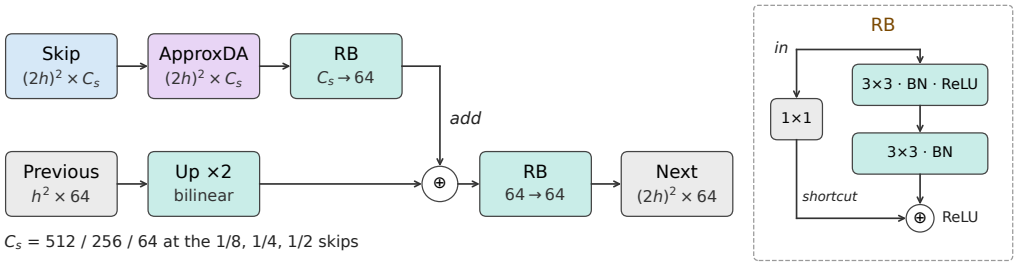


Figure 4: One slim decoder stage with additive skip fusion (64 channels). RB is the residual reduction block shown on the right.

where β is a learnable scalar initialized to 0. Groups are concatenated along the channel dimension to form $\mathbf{E}_{\text{CAM}} \in \mathbb{R}^{C' \times H \times W}$ and refined by a 3×3 convolution. When G does not divide C', the largest divisor of C' not exceeding G is used. Because C' is small in the lightweight model, $G = 8$ leaves a single channel per group in the bottleneck ($C' = 8$) and in the 1/2 skip ($C' = 4$, four groups), where CAM reduces to $(1+\beta)\mathbf{X}'$; the cost of CAM is negligible at every stage. $G = 8$ is therefore a fixed setting retained from the workshop design rather than a tuned choice; un-grouped CAM ($G = 1$) is evaluated in Section 5.

3.2.4. Fixed Equal-Blend Routing

In our earlier study [7], a learnable per-channel gate $\mathbf{g} = \sigma(\mathbf{W}_g \, \text{GAP}(\mathbf{x}))$ stayed at a near-uniform blend throughout training on Synapse (block means within 0.499–0.512 over 300 epochs) and did not outperform single-branch routing. ApproxDA-TransUNet therefore fixes the gate to an equal blend:

$$\mathbf{Y} = \mathbf{W}_f \, (0.5 \cdot \mathbf{E}_{\text{PAM}} + 0.5 \cdot \mathbf{E}_{\text{CAM}}) \in \mathbb{R}^{C \times H \times W}, \tag{9}$$

where $\mathbf{W}_f$ consists of channel dropout ($p = 0.05$), a 1×1 convolution back to C channels, and ReLU, as in the DANet head; there is no outer residual connection.

3.3. Compact Decoder Architecture

Unlike the concatenation decoder of DA-TransUNet ($256 \to 128 \to 64 \to 16$ channels), ApproxDA-TransUNet uses 64 channels at every skip stage and fuses skips by addition (Figure 4). A 3×3 convolution first maps $\mathbf{Z}$ to 64 channels. At each skip stage $l \in \{1, 2, 3\}$ (1/8, 1/4, 1/2), the previous map

Table 1: Parameters and FLOPs by component for DA-TransUNet and ApproxDA-TransUNet (224×224 input, 9 classes; fvcore on CPU, attention matrix products included). DA-TransUNet instantiates three DANet heads per decoder stage but executes one; its parameter count includes all of them.

Component	DA-TransUNet [6]		ApproxDA-TransUNet (Ours)	
	Params (M)	FLOPs (G)	Params (M)	FLOPs (G)
CNN stem	11.90 (R50, 3 stages)	3.96	1.45 (R50, 2 stages)	1.83
Token embedding	0.94	0.15	–	–
Bottleneck transformer	85.06 (ViT-B/16)	17.37	7.81 (LightViT-T)	0.75
Dual-attention blocks	2.68 (DANet heads)	0.87	0.54 (4 ApproxDA blocks)	0.60
Decoder (without attention blocks)	7.39	7.78	0.95	3.68
Segmentation head	< 0.01	0.07	< 0.01	0.07
Total	**107.95**	**30.21**	**10.75**	**6.92**
Relative	100%	100%	**10.0%**	**22.9%**

is bilinearly upsampled by $2\times$ and added to the ApproxDA-refined skip $\mathbf{S}_l$ after a channel-reduction block:

$$\mathbf{D}_l = \mathrm{RB}\left(\mathrm{Upsample}_{2\times}(\mathbf{D}_{l-1}) + \mathrm{RB}(\mathbf{S}_l)\right) \in \mathbb{R}^{64\times H_l \times W_l}, \tag{10}$$

where RB is a residual block of two 3×3 convolutions with batch normalization and a 1×1 shortcut. The full-resolution stage upsamples once more and applies two 3×3 convolutions with 16 channels, followed by a 1×1 segmentation head. The decoder without the skip blocks has 0.95M parameters (3.68 GFLOPs).

3.4. Computational Complexity and Parameter Breakdown

As detailed in Table 1, ApproxDA-TransUNet reduces the parameters from 107.95M to **10.75M** (-90.0%) and the FLOPs from 30.21G to **6.92G** (-77.1%). Most of the saving comes from replacing ViT-B/16 and from the slim decoder; the attention blocks themselves are a small part of either model. For high-resolution skip features (112×112, $N = 12{,}544$), windowing ($M = 28$) and low-rank projection ($r = 32$) cut spatial affinity entries from $N^2 = 1.57 \times 10^8$ to $N \cdot r = 4.01 \times 10^5$ (a $392\times$ reduction). Channel grouping divides the size of the CAM affinity matrix by G, but CAM is already negligible after the $C/16$ compression.

3.5. Optimization Objective

All models are trained with a composite objective combining Cross-Entropy ($\mathcal{L}_{\mathrm{CE}}$) and multi-class soft Dice loss ($\mathcal{L}_{\mathrm{Dice}}$) with equal weighting:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\mathrm{CE}} + 0.5 \cdot \mathcal{L}_{\mathrm{Dice}}. \tag{11}$$

No auxiliary losses or extra heads are used.

4. Experiments and Results

This section details the experimental protocol across five diverse clinical benchmarks, reports multi-seed quantitative comparisons between ApproxDA-TransUNet and DA-TransUNet, presents the FeTA fetal brain benchmark against 2D baselines and 3D references, and shows qualitative results.

4.1. Experimental Setup and Benchmark Datasets

Models are evaluated across **five clinical benchmarks** spanning four distinct imaging modalities under a strict **three-seed protocol** (seeds: 1234, 0, 1):

1. **Synapse Multi-Organ CT** [24]: 30 volumetric abdominal CT scans (18 train, 12 test; 2211 slices) covering 8 abdominal organs (aorta, gallbladder, left/right kidneys, liver, pancreas, spleen, stomach), trained for 300 epochs at 224×224 and evaluated with Dice (%) and HD95 in voxels (the TransUNet evaluation code, without voxel spacing).
2. **ACDC Cardiac Cine-MRI** [25]: 100 patient exams (40 end-diastolic/systolic 3D test volumes) targeting right ventricle, myocardium, and left ventricle, trained for 300 epochs at 224×224 and evaluated with Dice (%) and HD95 (px).
3. **CVC-ClinicDB Colonoscopy** [26]: 612 colonoscopy frames (75/25 split) for binary polyp segmentation, trained with Adam (10^{-3}) for 500 epochs at 256×256, evaluated with Dice (%) and HD95 (px).
4. **ISIC 2018 Dermoscopy** [27]: 2594 dermoscopic images (75/25 split) for binary skin-lesion segmentation, trained with SGD for 50 epochs at 256×256, evaluated with Dice (%) and HD95 (px).
5. **FeTA 2.4 Fetal Brain MRI** [28]: 80 reconstructed T2-weighted fetal brain MRI volumes under the standard EffiDec3D split (64 train, 8 val, 8 test) across 7 brain tissues (eCSF, GM, WM, LV, CBM, SGM, BS), trained for 61 epochs at 224×224 and evaluated with Dice (%) and HD95 (mm, using each subject's voxel spacing).

Implementation Protocols. All models are trained with the TransUNet augmentations (random 90° rotations with flips, and small rotations), a $0.5\,\mathcal{L}_{\text{CE}} + 0.5\,\mathcal{L}_{\text{Dice}}$ loss, batch size 24, and SGD (learning rate 0.01, momentum 0.9,

Table 2: Quantitative comparison across five clinical benchmarks evaluated over three independent random seeds (1234 / 0 / 1). DSC is reported as individual seed scores and Mean ± Standard Deviation. HD95 is reported in millimeters for FeTA, in voxels for Synapse, and in pixels for ACDC, CVC-ClinicDB, and ISIC 2018. p-values are paired t-tests over the three seeds of one dataset ($n = 3$, low power) and are given for completeness; see the text for the analysis across datasets.

Dataset / Model	Params (M)	FLOPs (G)	Dice (%) (Seed 1234 / 0 / 1)	Mean ± Std Dice (%)	HD95	Paired Diff. (vs. DA)
Synapse Multi-Organ CT (8 organs, HD95 in voxels)						
DA-TransUNet [6]	107.95	30.21	79.74 / 81.00 / 80.03	80.26 ± 0.66	**23.49**	–
ApproxDA-TransUNet (Ours)	**10.75**	**6.92**	80.61 / 80.75 / 80.36	**80.57 ± 0.20**	24.52	+**0.32** (p=0.43)
ACDC Cardiac Cine-MRI (3 structures, HD95 in px)						
DA-TransUNet [6]	107.95	30.21	90.66 / 91.06 / 91.09	90.94 ± 0.24	**1.52**	–
ApproxDA-TransUNet (Ours)	**10.75**	**6.92**	91.21 / 91.05 / 90.98	**91.08 ± 0.12**	1.91	+**0.15** (p=0.55)
CVC-ClinicDB Polyp (Binary polyp, HD95 in px)						
DA-TransUNet [6]	108.00	39.80	89.72 / 89.40 / 89.29	89.47 ± 0.23	14.49	–
ApproxDA-TransUNet (Ours)	**10.78**	**9.00**	90.32 / 89.37 / 89.94	**89.88 ± 0.48**	**12.74**	+**0.41** (p=0.20)
ISIC 2018 Dermoscopy (Binary skin lesion, HD95 in px)						
DA-TransUNet [6]	108.00	39.80	88.08 / 88.48 / 88.35	88.30 ± 0.20	**14.15**	–
ApproxDA-TransUNet (Ours)	**10.78**	**9.00**	88.43 / 88.82 / 88.27	**88.51 ± 0.28**	14.83	+**0.20** (p=0.29)
FeTA 2.4 Fetal Brain MRI (7 tissue classes, HD95 in mm)						
DA-TransUNet [6]	107.95	30.21	80.93 / 79.61 / 80.23	80.26 ± 0.66	**2.17**	–
ApproxDA-TransUNet (Ours)	**10.75**	**6.92**	81.66 / 81.08 / 80.41	**81.05 ± 0.63**	2.85	+**0.80** (p=0.17)
Combined Multi-Benchmark Aggregate (15 Seed Pairs Across 5 Modalities)						
Datasets with a higher mean: **5 / 5**; positive seed pairs: **10 / 15**						Average gain: +**0.37** pp
Parameter Reduction: **90.0%** (107.95M → 10.75M)						Compute Reduction: **77.1%** (30.21G → 6.92G)

weight decay 10^{-4}, polynomial decay), except CVC-ClinicDB (Adam, 10^{-3}). DA-TransUNet and ApproxDA-TransUNet are trained under identical settings. Every model is evaluated at the *final training epoch*, without checkpoint selection. Volume metrics are computed per case and averaged over classes; on FeTA, classes absent from a ground-truth volume are excluded from the average, as in nnU-Net.

4.2. Main Multi-Seed Benchmark Results

Table 2 summarizes the results. With a tenth of the parameters and under a quarter of the FLOPs, ApproxDA-TransUNet has a higher mean Dice than DA-TransUNet on all five benchmarks: +0.32 points on Synapse (seed standard deviation 0.20 vs. 0.66), +0.15 on ACDC, +0.41 on CVC-ClinicDB, +0.20 on ISIC 2018, and +0.80 on FeTA 2.4 (positive for all three seeds: +0.73, +1.47, +0.19). None of these per-dataset differences is significant with three seeds.

Across datasets, the three-seed mean is higher on all five datasets and 10 of the 15 seed pairs favor ApproxDA-TransUNet, with an average gain of +0.37 **points**. Because the three seeds of a dataset are not independent experiments, we treat the five datasets as the independent experimental units:

Table 3: Comparison with published results. Upper block: values reported in the literature (single runs, original training settings); Synapse, CVC-ClinicDB, and ISIC 2018 as compiled in the DA-TransUNet paper [6], ACDC from the Swin-Unet paper [11]. Lower block: models retrained in this work (three-seed means, final epoch). Synapse and ACDC use the common TransUNet test sets (12 CT volumes; 70/10/20 ACDC patients); our CVC-ClinicDB and ISIC 2018 splits (75/25) differ from those of the literature, so these columns are indicative only. HD95 in voxels; the best value in each column is in bold.

Method	**Synapse**		**ACDC**	**CVC-ClinicDB**		**ISIC 2018**	
	DSC (%)	HD95	DSC (%)	DSC (%)	IoU (%)	DSC (%)	IoU (%)
U-Net [1]	76.85	39.70	87.55†	86.93	78.21	87.22	81.14
Attention U-Net [4]	77.77	36.02	86.75†	87.41	79.35	87.60	81.51
U-Net++ [2]	76.91	36.93	–	87.14	78.47	87.30	81.33
TransUNet [3]	77.48	31.69	89.71	89.01	81.63	88.78	82.63
Swin-Unet [11]	79.13	**21.55**	90.00	–	–	–	–
DA-TransUNet [6]	79.80	23.48	–	89.47	82.51	**88.88**	**82.78**
DA-TransUNet [6] (retrained)	80.26	23.49	90.94	89.47	84.55	88.30	81.54
ApproxDA-TransUNet (Ours)	**80.57**	24.52	**91.08**	**89.88**	**85.19**	88.51	81.63

†ResNet-50 encoder variants (R50 U-Net, R50 Att-UNet) as reported by Cao et al. [11].

the mean dataset-level gain is +0.37 points (one-sample t-test across the five dataset means, $p = 0.031$), and a hierarchical bootstrap that resamples datasets and then seeds yields a 95% interval of $[0.13, 0.68]$ points. With only five datasets these estimates are themselves uncertain, and we read them as evidence that accuracy is retained rather than as a claim of superiority. HD95 does not follow the Dice trend: it is lower than DA-TransUNet's only on CVC-ClinicDB (12.74 vs. 14.49 px) and higher on the other four datasets (significantly only on FeTA, $p = 0.04$, see the case audit below).

4.3. Comparison with Published Results

Table 3 relates the results to published numbers. On the two benchmarks with shared test sets, ApproxDA-TransUNet is above all reported values: 80.57% on Synapse against 79.80% for DA-TransUNet and 79.13% for Swin-Unet, and 91.08% on ACDC against 90.00% for Swin-Unet. Our retrained DA-TransUNet reaches 80.26% on Synapse, close to the reported 79.80%, which supports the fairness of our training setup. On ISIC 2018 our retrained models are slightly below the reported values, which were obtained on a different split.

4.4. FeTA 2.4 Benchmark: 2D and 3D Comparisons

Table 4 places ApproxDA-TransUNet among six 2D models trained under the same protocol and three 3D references:

Table 4: Tissue-level Dice (%) on the FeTA 2.4 test set (8 subjects). 2D models: three seeds under the protocol of Section 4, last epoch; per-tissue values are three-seed means, and p is a paired t-test against ApproxDA-TransUNet. FLOPs are per 224×224 slice. 3D models: our reproduction under the EffiDec3D protocol, one seed, checkpoint selected on the validation set; per-tissue values were not recorded, and their cost is given per 96^3 patch in multiply–accumulates (GMac), so it is not directly comparable with the 2D FLOPs.

Model	Type	Params (M)	Cost	eCSF	GM	WM	LV	CBM	SGM	BS	Mean DSC (%)	p vs. ours
U-Net [1]	2D	31.04	36.9 GFLOPs	73.8	67.7	84.4	84.1	**82.5**	83.1	78.9	79.21 ± 0.34	0.018
Attention U-Net [4]	2D	31.39	37.7 GFLOPs	73.5	66.8	84.5	84.4	82.2	83.5	78.9	79.11 ± 0.34	0.073
U-Net++ [2]	2D	9.16	26.5 GFLOPs	73.3	67.4	84.1	83.8	82.3	83.3	78.9	79.01 ± 0.16	0.021
Swin-UNet [11]	2D	27.17	6.2 GFLOPs	74.1	68.6	83.9	83.6	81.1	84.7	74.0	78.60 ± 0.46	0.059
TransUNet [3]	2D	105.28	29.3 GFLOPs	74.5	68.0	87.7	**88.9**	82.2	**85.0**	**81.4**	**81.12 ± 0.63**	0.036
DA-TransUNet [6]	2D	107.95	30.2 GFLOPs	73.9	68.7	85.2	87.4	81.9	84.1	80.6	80.26 ± 0.66	0.17
ApproxDA (Ours)	2D	**10.75**	6.9 GFLOPs	**75.5**	**71.3**	**87.9**	87.9	81.8	83.7	79.3	81.05 ± 0.63	–
3D reference models (our reproduction, EffiDec3D split, one seed, validation-selected checkpoint)												
SwinUNETR [13]	3D	62.19	308.0 GMac	–	–	–	–	–	–	–	78.38	–
3D UX-Net [29]	3D	53.01	578.5 GMac	–	–	–	–	–	–	–	80.01	–
MedNeXt-M [30]	3D	17.56	106.1 GMac	–	–	–	–	–	–	–	81.22	–

- **CNN and Swin baselines**: ApproxDA-TransUNet (81.05%) is above U-Net (+1.85 points, $p = 0.018$), U-Net++ (+2.04, $p = 0.021$), Attention U-Net (+1.94, $p = 0.073$), and Swin-UNet (+2.46, $p = 0.059$).
- **Transformer hybrids**: it is 0.07 points below TransUNet, a small but consistent gap (lower for all three seeds, $p = 0.036$), with a tenth of the parameters and under a quarter of the FLOPs, and 0.80 points above DA-TransUNet ($p = 0.17$).
- **3D references**: its mean Dice lies between 3D UX-Net (80.01%) and MedNeXt-M (81.22%). The 3D numbers come from a single seed with validation-based checkpoint selection, so this comparison is indicative only.
- **Tissues**: relative to DA-TransUNet, the gains are in grey matter (+2.6), white matter (+2.7), and external CSF (+1.6); cerebellum and deep grey matter are on par (−0.1, −0.4), and the brainstem is lower (−1.3).

Case Audit of FeTA HD95 (Sub-052 Outlier). ApproxDA-TransUNet has a higher mean HD95 on FeTA than DA-TransUNet (2.85 vs. 2.17 mm, Table 2). The difference comes from a single subject, **sub-052**, which has severe ventriculomegaly (lateral-ventricle to white-matter volume ratio 2.29, against a training median of 0.16 and a training maximum of 1.81) and

is therefore outside the training distribution. On this subject, ApproxDA-TransUNet predicts a few small cerebellum regions (the largest has 34, 170, and 266 voxels for the three seeds) about 50 mm away from the true cerebellum, which raises the cerebellum HD95 to 28–50 mm and the subject's mean HD95 to 8.88, 8.89, and 13.65 mm (DA-TransUNet: 2.85, 4.43, 4.02 mm). Excluding sub-052, the mean HD95 over the other seven subjects is 1.76 **mm** for ApproxDA-TransUNet and 1.94 mm for DA-TransUNet.

4.5. Qualitative Visual Evaluation

Figure 5 shows one representative example per dataset. On these median cases the two models produce similar masks: ApproxDA-TransUNet is slightly better on ACDC, ISIC 2018, and FeTA 2.4 and slightly worse on Synapse and CVC-ClinicDB. The visible differences are small and lie mostly along structure boundaries.

5. Ablation Studies

Single-axis ablations are run on Synapse with three seeds (1234, 0, 1). The reference is ApproxDA-TransUNet ($80.57 \pm 0.19\%$ Dice, HD95 24.52 voxels); each variant changes exactly one factor.

5.1. Dual-Attention Blocks and Their Approximation

- **The blocks matter in this configuration**: removing all ApproxDA blocks lowers Dice on Synapse by 2.76 **points** for every seed ($p = 0.008$) and raises HD95 from 24.52 to 29.12; the largest drops are in the gallbladder and pancreas.

- **The approximation costs no accuracy**: replacing the approximated attention by the exact global dual attention of DA-TransUNet ($M = 112$, $r = 0$, $G = 1$) leaves Dice unchanged (-0.01, $p = 0.97$), and exact attention inside each window ($r = 0$) is also on par ($+0.19$). The exact global variant costs 14% more FLOPs and about twice the training time per run (94 vs. 45 minutes).

5.2. Approximation Hyperparameters

- **Window size and rank**: a global window ($M = 112$, -0.50, $p = 0.54$) and a higher rank ($r = 64$, -0.93, $p = 0.11$) are not significantly different from the defaults; with three seeds on Synapse, differences below about one point cannot be resolved.

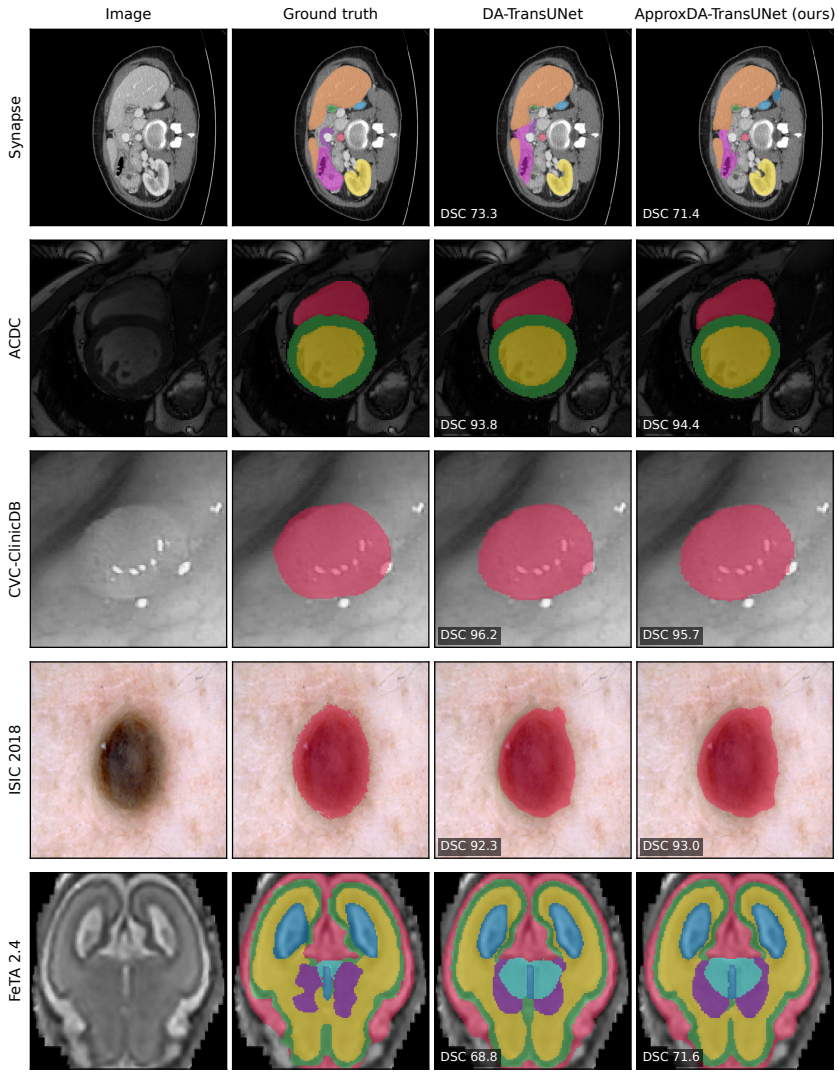


Figure 5: Qualitative results (seed 1234, final epoch). For each dataset, the example is chosen without reference to DA-TransUNet: the test case with the median Dice of ApproxDA-TransUNet and, for volumes, the slice with the most annotated classes. Panels are cropped to the annotated region; the Dice of the displayed slice or image is given on each prediction.

Table 5: Single-axis ablations on Synapse (seeds 1234 / 0 / 1, 224×224, 9 classes). The baseline is ApproxDA-TransUNet ($M = 28$, $r = 32$, $G = 8$, $g = 0.5$, LightViT-T, slim decoder). HD95 in voxels. Paired differences and p-values (paired t-test over three seeds, $n = 3$) are relative to the baseline. FLOPs are counted on CPU with explicit attention products.

Ablation Variant	Params (M)	FLOPs (G)	Mean DSC (%)	HD95	Paired Diff. (p)
ApproxDA-TransUNet baseline	**10.75**	**6.92**	**80.57 ± 0.19**	**24.52**	–
Block presence and exact dual attention					
w/o ApproxDA blocks	10.20	6.3	77.82 ± 0.33	29.12	−2.76 (0.008)
Exact global dual attention (M=112, r=0, G=1)	10.65	7.90	80.56 ± 0.62	23.64	−0.01 (0.97)
Window size M					
$M = 112$ (global window at every stage)	12.25	6.92	80.07 ± 0.98	28.99	−0.50 (0.54)
Projection rank r					
$r = 0$ (exact attention inside each window)	10.65	7.03	80.76 ± 0.15	21.12	+0.19 (0.44)
$r = 64$	10.85	6.93	79.64 ± 0.40	26.85	−0.93 (0.11)
Channel groups G					
$G = 1$ (un-grouped CAM)	10.75	6.92	81.12 ± 0.77	23.48	+0.55 (0.25)
$G = 4$	10.75	6.92	79.91 ± 0.57	26.69	−0.67 (0.10)
Encoder					
R50 third stage instead of LightViT-T	15.42	8.7	78.81 ± 0.27	27.71	−1.76 (0.007)
ViT-B/16 instead of LightViT-T	100.47	26.0	78.52 ± 1.64	26.01	−2.05 (0.14)
LightViT-S instead of LightViT-T	20.50	8.0	80.31 ± 0.63	23.62	−0.26 (0.43)
Bottleneck placement and decoder					
Bottleneck block before LightViT-T	11.06	7.2	79.47 ± 0.36	26.10	−1.10 (0.008)
Full decoder ($256 \to 128 \to 64 \to 16$)	14.24	10.4	79.60 ± 1.64	25.63	−0.97 (0.40)

- **Channel groups**: $G = 8$ is a fixed default carried over from the ViT-B/16 version of the block, where the bottleneck has $C' = 48$ channels and each group six; it was not tuned for the lightweight model. Here it leaves one channel per group in the bottleneck ($C' = 8$) and the 1/2 skip ($C' = 4$, four groups), where CAM degenerates to the scaling $(1 + \beta)\mathbf{X}'$ (Section 3). Un-grouped CAM ($G = 1$) is 0.55 points higher ($p = 0.25$) at no extra cost. We nevertheless keep $G = 8$ in the final model as a fixed, non-tuned setting, for consistency with the workshop design and with all other experiments reported here; the results do not show that $G = 8$ is optimal, and $G = 1$ or a stage-adaptive grouping (Section 6) is a reasonable alternative.
- **Channel groups** $G = 4$: -0.67 ($p = 0.10$), not significant.
- **Encoder**: replacing LightViT-T with the third ResNet-50 stage (the TransUNet stem, 15.42M parameters) lowers Dice by 1.76 points for every seed ($p = 0.007$) while adding 4.7M parameters. ViT-B/16 is also lower (-2.05, $p = 0.14$), and the larger LightViT-S is not better (-0.26, $p = 0.43$), so the lightest transformer tested is sufficient.

- **Block placement**: applying the bottleneck block before LightViT-T, on the $28 \times 28 \times 512$ stage-2 output as in DA-TransUNet's ordering, lowers Dice by 1.10 points for every seed ($p = 0.008$) and adds 0.31M parameters. The full decoder of DA-TransUNet is not better than the slim decoder (-0.97, $p = 0.40$) at 1.3 times the parameters and 1.5 times the FLOPs.

6. Discussion and Mechanistic Insights

This section discusses left–right errors caused by mirror augmentation, reports deployment costs, and outlines limitations.

6.1. Anatomical Symmetries and Data Augmentation

In an error analysis on the Synapse test set (with the earlier ViT-B/16 model), most misclassified voxels were boundary or extent errors between an organ and the background. The confusions between organs were mostly whole-organ misidentifications, chiefly stomach versus spleen and left versus right kidney, and were concentrated in a few test cases.

The left–right confusions are a plausible side effect of the standard TransUNet augmentation: `random_rot_flip` follows a random 90° rotation with an axis flip but does not swap the left and right labels, so about half of the augmented slices show mirrored anatomy with unchanged laterality labels. We did not quantify this effect on the reported benchmarks; laterality-aware augmentation, which swaps the labels of bilateral structures when an image is mirrored, is a simple safeguard for such structures.

6.2. Computational Efficiency and Deployment Profile

Table 6 summarizes the deployment cost. ApproxDA-TransUNet needs 58% less GPU memory for training and inference and stores a ten-times smaller checkpoint. Because volumes are segmented slice by slice in batches, throughput is the relevant speed measure, and it is 2.55 times higher. At batch size 1, however, the lightweight model is slower (9.6 vs. 6.5 ms): its many small LightViT and attention kernels leave the GPU underutilized, so the FLOP reduction does not translate into lower single-slice latency.

Table 6: Deployment profile on an NVIDIA RTX 5090 (fp32, 224×224 input, 9 classes, idle GPU). Memory is the peak allocated by PyTorch, including weights and cuDNN workspace; training uses batch size 24 with SGD.

Metric	DA-TransUNet	ApproxDA-TransUNet (Ours)
Parameters	107.95 M	**10.75 M** (−90.0%)
GFLOPs per slice	30.21	**6.92** (−77.1%)
Checkpoint size (FP32)	432 MB	**43.5 MB**
Peak training memory (batch 24)	11.52 GiB	**4.85 GiB** (−58%)
Peak inference memory (batch 1)	2.59 GiB	**1.09 GiB** (−58%)
Throughput, batch 32 (slices/s)	555	**1415** (2.55×)
Latency, batch 1	**6.5 ms**	9.6 ms

6.3. Limitations and Future Work

The channel grouping is a fixed default ($G = 8$) rather than an optimized design: with the small compressed widths of the lightweight model it removes cross-channel interaction in two of the four blocks. A stage-adaptive grouping that keeps at least two channels per group (for example $G = C'/4$) would be the more principled choice. The ablations were run on Synapse only, where three seeds resolve differences of about one Dice point at best, and the statistical analysis across datasets rests on five benchmarks. ApproxDA-TransUNet processes 3D scans on a 2D planar basis. It therefore cannot use inter-slice context, which matters, for example, at the cranial and caudal ends of organs. Extending ApproxDA blocks to lightweight 2.5D or anisotropic 3D formulations presents a promising avenue for volumetric research.

7. Conclusion

In this paper, we presented **ApproxDA-TransUNet**, a lightweight redesign of DA-TransUNet that combines a truncated ResNet-50 stem, LightViT-T, approximated dual-attention blocks (windows of $M = 28$, rank $r = 32$, and a fixed, non-tuned channel grouping) on the bottleneck and all skips, and a slim decoder. It uses 90.0% fewer parameters (10.75M vs. 107.95M) and 77.1% fewer FLOPs (6.92 vs. 30.21 GFLOPs). Over three seeds on five benchmarks (Synapse, ACDC, CVC-ClinicDB, ISIC 2018, and FeTA 2.4), its Dice is comparable to that of DA-TransUNet (higher three-seed mean on all five datasets, average gain +0.37 points), while HD95 is higher on four of the five datasets; on FeTA 2.4 it is within 0.1 points of TransUNet. In deployment it needs 58% less GPU memory and processes 2.55 times more

slices per second at batch size 32, although single-slice latency at batch size 1 is not lower.

In single-axis ablations on Synapse, removing the dual-attention blocks reduces Dice by 2.76 points, and windowed low-rank position attention and grouped channel attention match the exact global dual attention of DA-TransUNet at a fraction of its cost, showing that, in the evaluated configuration, dual attention can be retained without its full quadratic overhead.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT Authorship Contribution Statement

Author One: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - Original Draft, Visualization. **Author Two**: Software, Validation, Data curation, Writing - Review & Editing. **Author Three**: Supervision, Project administration, Resources, Funding acquisition, Writing - Review & Editing.

Data and Code Availability

All datasets used in this study (Synapse BTCV, ACDC, CVC-ClinicDB, ISIC 2018, and FeTA 2.4) are publicly accessible benchmark repositories. The complete source code and pretrained model checkpoints for ApproxDA-TransUNet will be made publicly available upon publication.

References

[1] O. Ronneberger, P. Fischer, T. Brox, U-Net: Convolutional networks for biomedical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), Springer, 2015, pp. 234–241.

[2] Z. Zhou, M. M. Rahman Siddiquee, N. Tajbakhsh, J. Liang, UNet++: A nested U-Net architecture for medical image segmentation, in: International Workshop on Deep Learning in Medical Image Analysis, Springer, 2018, pp. 3–11.

[3] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, Y. Zhou, TransUNet: Transformers make strong encoders for medical image segmentation, arXiv preprint arXiv:2102.04306 (2021).

[4] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz, et al., Attention U-Net: Learning where to look for the pancreas, arXiv preprint arXiv:1804.03999 (2018).

[5] J. Fu, J. Liu, H. Tian, Y. Li, Y. Bao, Z. Fang, H. Lu, Dual attention network for scene segmentation, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, 2019, pp. 3141–3149.

[6] G. Sun, Y. Pan, W. Kong, Z. Xu, J. Ma, T. Racharak, L.-M. Nguyen, J. Xin, DA-TransUNet: Integrating spatial and channel dual attention with transformer U-Net for medical image segmentation, Frontiers in Bioengineering and Biotechnology 12 (2024) 1398237.

[7] J. Zhong, Q. Chen, H. Zhang, ApproxDA-TransUNet: Exploring window sensitivity in dual-attention approximation for medical image segmentation, in: IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2026.

[8] T. Huang, Z. Lang, L. Yuan, J. Feng, LightViT: Towards light-weight vision transformer with inherited top-down interaction, arXiv preprint arXiv:2207.05534 (2022).

[9] N. Ibtehaz, M. S. Rahman, MultiResUNet: Rethinking the U-Net architecture for multimodal biomedical image segmentation, Neural Networks 121 (2020) 74–87.

[10] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby, An image is worth 16×16 words: Transformers for image recognition at scale, in: International Conference on Learning Representations (ICLR), 2021.

[11] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, M. Wang, Swin-Unet: Unet-like pure transformer for medical image segmentation, in: European Conference on Computer Vision (ECCV Workshops), Springer, 2022, pp. 205–218.

[12] A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H. R. Roth, D. Xu, UNETR: Transformers for 3D medical image segmentation, in: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2022, pp. 574–584.

[13] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, D. Xu, Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images, arXiv preprint arXiv:2201.01266 (2022).

[14] R. Azad, R. Arimond, E. K. Aghdam, A. Kazerouni, D. Merhof, DAE-Former: Dual attention-guided efficient transformer for medical image segmentation, in: International Workshop on Predictive Intelligence in Medicine, Springer, 2023, pp. 83–95.

[15] B. Azad, P. Adibfar, K. Fu, TransDAE: Dual attention mechanism in a hierarchical transformer for efficient medical image segmentation, arXiv preprint arXiv:2409.02018 (2024).

[16] G. Ding, J. Wu, K. Yang, Z. Huang, Z. Chen, DAMamba: Dual-attention mamba for lightweight and robust breast ultrasound tumor segmentation, in: IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2025, pp. 5416–5423.

[17] S. Zhu, Y. Yan, L. Wei, Y. Li, T. Mao, X. Dai, R. Du, SECA-Net: Squeezed-and-excitated contextual attention network for medical image segmentation, Biomedical Signal Processing and Control 97 (2024) 106704. doi:10.1016/j.bspc.2024.106704.

[18] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin Transformer: Hierarchical vision transformer using shifted windows, in: IEEE/CVF International Conference on Computer Vision (ICCV), IEEE, 2021, pp. 9992–10002.

[19] S. Wang, B. Z. Li, M. Khabsa, H. Fang, H. Ma, Linformer: Self-attention with linear complexity, arXiv preprint arXiv:2006.04768 (2020).

[20] K. Choromanski, V. Likhosherstov, D. Dohan, X. Song, A. Gane, T. Sarlos, P. Hawkins, J. Davis, A. Mohiuddin, L. Kaiser, et al., Rethinking attention with performers, arXiv preprint arXiv:2009.14794 (2020).

[21] M. M. Rahman, M. Munir, R. Marculescu, EMCAD: Efficient multi-scale convolutional attention decoding for medical image segmentation, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, pp. 11769–11779.

[22] Y. Chen, B. Ma, Y. Xia, Adaptive decoder-block selection with filter reweighting for medical image segmentation, Biomedical Signal Processing and Control 85 (2023) 105033. doi:10.1016/j.bspc.2023.105033.

[23] K. Peng, B. Hong, Y. Liu, C. Chen, Y. Pan, Y. Jiang, X. Liu, FE-MAANet: A frequency-enhanced multi-scale adaptive attention network for medical image segmentation, Biomedical Signal Processing and Control 113 (2026) 108884. doi:10.1016/j.bspc.2025.108884.

[24] B. Landman, Z. Xu, J. E. Iglesias, M. Styner, T. Langerak, A. Klein, MICCAI multi-atlas labeling beyond the cranial vault—workshop and challenge, in: Proc. MICCAI Multi-Atlas Labeling Beyond Cranial Vault Workshop Challenge, Vol. 5, 2015, p. 12.

[25] O. Bernard, A. Lalande, C. Zotti, F. Cervenansky, J. Yang, P.-A. Heng, I. Cetin, G. Sanroma, M. Peterson, C. Radford, et al., Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: Is the problem solved?, IEEE Transactions on Medical Imaging 37 (11) (2018) 2514–2525.

[26] J. Bernal, F. J. Sánchez, G. Fernández-Esparrach, D. Gil, C. Rodríguez, F. Vilariño, WM-DOVA maps for accurate polyp highlighting in colonoscopy: Validation vs. saliency maps from physicians, Computerized Medical Imaging and Graphics 43 (2015) 99–111.

[27] N. Codella, V. Rotemberg, P. Tschandl, M. E. Celebi, S. Dusza, D. Gutman, B. Helba, A. Kalloo, K. Liopyris, M. Marchetti, et al., Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (ISIC), arXiv preprint arXiv:1902.03368 (2019).

[28] K. Payette, P. de Dumast, H. Kebiri, I. Ezhov, J. C. Paetzold, S. Shit, A. Iqbal, R. Khan, R. Kottke, P. Grehten, et al., An automatic multi-tissue human fetal brain segmentation benchmark using the Fetal Tissue Annotation Dataset, Scientific Data 8 (1) (2021) 167.

[29] H. H. Lee, S. Bao, Y. Huo, B. A. Landman, 3D UX-Net: A large kernel volumetric ConvNet modernizing hierarchical Transformer for medical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), Springer, 2022, pp. 703–713.

[30] S. Roy, G. Koehler, C. Ulrich, M. Baumgartner, J. Petersen, F. Isensee, P. F. Jaeger, K. H. Maier-Hein, MedNeXt: Transformer-driven scaling of ConvNets for medical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), Springer, 2023, pp. 405–415.